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Container Orchestration Storage › **Volumes**

Container Orchestration Storage

Volumes

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This article provides a guide on persistent volumes in Kubernetes, explaining their importance for data persistence beyond a pod’s lifecycle.

Hello and welcome to our comprehensive guide on persistent volumes in Kubernetes. In this article, we’ll begin by examining how volumes work in Docker, then transition into exploring their implementation in Kubernetes to ensure data persistence beyond a pod’s lifecycle.

Volumes in Docker

Docker containers are inherently transient—they are designed to run temporarily, process data, and then be destroyed. By default, any data generated within a container is lost once the container stops. To overcome this limitation, Docker allows you to attach a volume at container creation. This attached volume ensures that data persists even after the container is terminated.

Similarly, Kubernetes pods are ephemeral. When a pod processes data and is eventually deleted, any data stored within it is lost unless a volume is attached. Volumes in Kubernetes ensure that essential data remains available even after a pod’s lifecycle ends.

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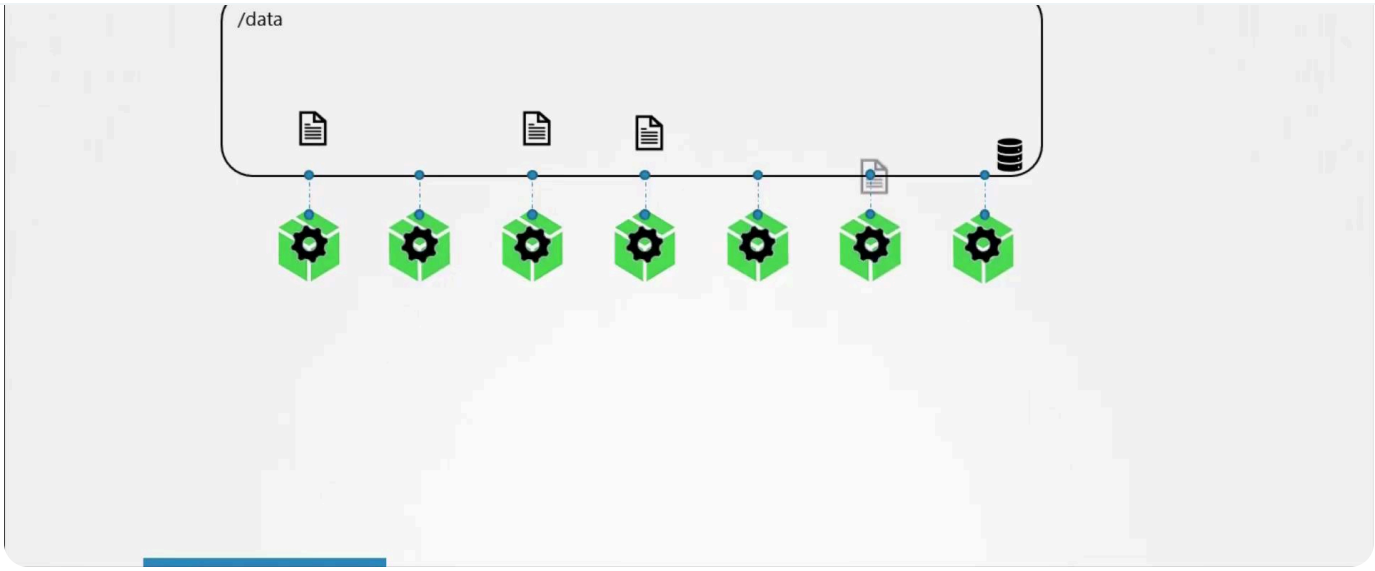

A Simple Volume Implementation in Kubernetes

Consider a simple example on a single-node Kubernetes cluster. In this scenario, a pod generates a random number between 0 and 100 and writes it to `/opt/number.out`. Without a volume, this file would be lost when the pod is deleted. To retain the generated number, we create a volume and mount it into the pod.

In this example, we use a directory on the host as our storage medium. The volume is configured to use the `/data` directory on the node and is mounted to the `/opt` directory inside the container. This ensures that any data written to `/opt/number.out` is persisted to the host directory.

Below is the YAML configuration for our pod with a hostPath volume:

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```
spec:
  containers:
  - image: alpine
    name: alpine
    command: ["/bin/sh", "-c"]
    args: ["shuf -i 0-100 -n 1 >> /opt/number.out;"]
    volumeMounts:
    - mountPath: /opt
      name: data-volume
  volumes:
  - name: data-volume
    hostPath:
      path: /data
      type: Directory
```

In this configuration:

- The container runs an Alpine Linux image that executes the `shuf` command to generate a random number and appends the result to the `/opt/number.out` file.
- The `volumeMounts` field ensures that the `data-volume` is mounted at `/opt` within the container.
- The `volumes` section defines the `data-volume` using the host's `/data` directory.

> Even if the pod is deleted, the file containing the random number remains stored on the host, preserving your generated data.

Considerations for Multi-Node Clusters

While the hostPath volume works well in a single-node environment, it is not recommended for multi-node Kubernetes clusters. In a multi-node setup, pods scheduled on different nodes would reference their local `/data` directories, leading to inconsistent data storage.

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- Network File System (NFS)
- GlusterFS
- Flocker
- Fibre Channel
- CephFS
- ScaleIO

as well as public cloud storage options like:

- AWS EBS
- Azure Disk or File storage
- Google Persistent Disk

For example, to configure an AWS Elastic Block Store volume instead of a hostPath, update the volume configuration as follows:

```
volumes:
  - name: data-volume
    awsElasticBlockStore:
      volumeID: <volume-id>
      fsType: ext4
```

In this updated configuration, the `awsElasticBlockStore` field is specified along with the volume ID and file system type (ext4). This setup enables Kubernetes to manage volume storage on AWS EBS, providing a scalable and reliable external storage solution.

> Using hostPath in multi-node clusters can lead to data inconsistency. Always consider using external storage solutions for environments that require shared storage across nodes.

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